

Zoe Watson (they/them)

Solutions Architecture · Software & Data Engineering · Research & Analysis

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KEY ACHIEVEMENTS

Sony Imageworks Innovation Award — for the greenfield rebuild of a legacy tool into a secure cross-platform app: 50+ features, 2 languages, 300+ users, in under 8 months. At the same time taught a new tool stack to 50+ colleagues, from 1-on-1s to department-wide sessions, directly enabling multiple follow-on projects.

Peer-reviewed publication — climate model linking collective community action to global emissions outcomes, The Expedition UBC 2023.

Security clearance — held an active clearance while overseeing the technical security and integrity of Personally Identifiable Information (PII) (2018–2020).

PROJECTS

Founder — SparkForge Workshop

Jan 2023 – Present

- Founded and independently operate a small craft business—managing product development, supply chain, financial accounting, web development, and market operations.
- Grew gross revenue 928.6% YoY in 2025; on pace for 50% growth in 2026.
- Cut production cost from 34% to 25% of sell price via unit-economics analysis.
- Built the e-commerce site, driving 27% YoY traffic growth.

EDUCATION

University of British Columbia

BSc, Combined Major in Science: Physics, Computer Science & Environmental Science (2024)

VOLUNTEERING

- **Robotics Coach, FIRST Robotics** — mentored 40+ students in programming and robotics software.
- **Greater Victoria Green Team** — local ecosystem restoration and invasive species removal.

LANGUAGES

English (expert) · French (intermediate)

WORK AUTHORIZATION

Canada - Unrestricted Work Permit and Permanent Resident candidate.

United States - Citizen.

New Zealand - Permanent Resident.

SUMMARY

Solutions-minded engineer with 4+ years spanning software development, data architecture, and business intelligence. I take complex problems from idea to impact: scoping with scientists, executives, and end users; architecting the data and software; and delivering and tools that make an impact.

WORK EXPERIENCE

Research Developer — University of Victoria

Feb 2026 – Present

- Leading design of a climate mitigation simulation tool for small and mid-size municipalities, coordinating technical requirements across academic science leads and local-government stakeholders.
- Designed and built multi-source data pipelines (Python, SQL) ingesting 8,400+ records across 13 UN Decades from 20+ web sources, owning the full lifecycle from extraction through cleaning, transformation, and structured storage.
- Architected the dataset's semantic layer from scratch — entities, relationships, unit taxonomies, and cross-dataset metric definitions — unlocking downstream analysis and topic modeling.
- Built and iterated an LLM agent workflow for data extraction and cleaning, replacing manual processes with a repeatable, scalable pipeline.
- Applied panel statistical analysis (Granger causality, Dumitrescu-Hurlin) across 151 countries and 272 variable pairs; surfaced systemic patterns stakeholders used immediately to draft a policy rebuttal.

Full-Stack Software Developer — CityView

Jan 2025 – Jun 2026

- Delivered city-management software to municipal governments across Canada and the U.S. on a collaborative, cross-functional team.
- Spearheaded the redesign and relaunch of a core product dashboard — migrating C# to TypeScript and Blazor — resolving a major customer pain point with a modernized UI, modular widget architecture, and advanced data visualization.
- Partnered with QA and Sales to align technical priorities with customer needs, driving product decisions from user feedback and business objectives.

Software Engineer — Sony Pictures Imageworks

Aug 2022 – Apr 2023

- Owned a product end-to-end: ran the feasibility study and tech-stack selection, self-taught Electron to a first deliverable in 2 weeks, and shipped the award-winning application on schedule.
- Led user research across 5 departments and 3 department heads; made the case for pre-release user feedback, surfacing critical features the team had not considered.
- Collaborated with senior non-technical stakeholders on a design roadmap balancing user needs against budget and time constraints.

Software Developer — Motorola Solutions

Sep 2021 – May 2022

- Developed features for Avigilon Control Center, an enterprise security-camera SaaS, in C++/C#, React, and TypeScript, from visual design through unit tests in a CI/CD pipeline.
- Led backlog meetings, assessed feature cost and risk in sprint planning, and presented to stakeholders in sprint reviews.

Information Technology Developer — Intellectus, LLC

May 2018 – Jul 2020

- Stepped in after the head of IT retired, owning servers, security compliance, and stakeholder requirements across three consecutive summers.
- Designed a digital onboarding and pre-employment testing platform that processed 200+ new hires; built and rolled out a security compliance upgrade to 350+ workstations.

Zoe Watson — Skills & Demonstrations

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PROGRAMMING LANGUAGES

Object Oriented: C, C#, C++, Java, Python, JavaScript (React, Node, Typescript, jQuery, AngularJS, Three JS, WebGL)

Web Development: CSS, HTML, PHP, JSON

Databases: MySQL, SQL, UML, XML

Unix / Linux: Shell, Bash, Git Bash

Production code in 5+ languages

Shipped production software across four companies — from C++/C# enterprise security at Motorola to a TypeScript/Blazor dashboard relaunch at CityView.

DATA ENGINEERING & ANALYSIS

Data pipeline architecture, web scraping & extraction, MySQL/SQL, semantic data architecture & taxonomy design, data governance & quality standards, statistical analysis (panel Granger causality, Dumitrescu-Hurlin), data visualization & analytical reporting

8,400+ records, 151 countries

Built multi-source pipelines and the full semantic layer at UVic; ran panel statistical analysis across 272 variable pairs feeding a policy rebuttal.

AI & LLM SYSTEMS

LLM agent workflow design, statistical modeling, NLP, machine learning & deep learning fundamentals, AI regulation & ethics

Manual process → scalable pipeline

Designed and iterated an LLM agent workflow for data extraction and cleaning, replacing manual processes with a repeatable architecture.

FRAMEWORKS, CLOUD & DEVOPS

Electron, .NET / .NET Core, Blazor, Symfony, Docker, Kubernetes, Azure, AWS, Git/GitHub/GitLab, CI/CD, automated & unit testing, infrastructure automation, SOLID/TDD best practices

Self-taught to shipped in 2 weeks

Self-taught Electron to a first deliverable in 2 weeks at Sony; migrated StrataPress from SFTP to Git and built its CI/CD infrastructure.

DOMAIN KNOWLEDGE

Climate science & mitigation, climate modeling & scenario analysis, carbon accounting frameworks (GHG Protocol, SBTi, TCFD, GRI, SASB), environmental systems (water, energy, food, waste), urban geography & planning, housing policy, ESG frameworks & reporting

Climate tools for real decisions

Leading design of a climate mitigation simulation tool for small and mid-size municipalities, bridging academic science leads and local government.

RESEARCH & EVIDENCE SYNTHESIS

Research design & taxonomy development, scientific literature synthesis, evidence synthesis, quantitative & qualitative research methods, policy-facing research reporting, methodology documentation & robustness checks, peer-reviewed publishing

Peer-reviewed at UBC

Published a climate action model in The Expedition UBC; architected UVic's research taxonomy and authored a documented statistical report with robustness checks.

SECURITY & COMPLIANCE

Security clearance experience, PII protection & data privacy, security compliance auditing & remediation, secure application development, server administration & backup integrity

Cleared and trusted

Held an active security clearance while safeguarding PII (2018–2020); built and rolled out a security compliance upgrade to 350+ workstations.

PROJECT & PRODUCT MANAGEMENT

Agile (Scrum, Kanban, sprint planning, code review), roadmap design & management, user research & stakeholder interviews, requirements gathering, risk & cost assessment, legacy system modernization, end-to-end product delivery

Idea to 300+ users in 8 months

Took SPNote from feasibility study through roadmap, user research across 5 departments, and delivery — earning an Innovation Award.

BUSINESS INTELLIGENCE & STRATEGY

Unit economics & COGS optimization, revenue growth strategy, market development, consultative sales, e-commerce management, financial accounting, supply chain management

928.6% revenue growth

Sole operator of SparkForge Workshop: cut COGS from 34% to 25%, expanded from 2 to 11 markets, grew web traffic 27% YoY.

COMMUNICATION & LEADERSHIP

Knowledge mobilization, technical translation for non-technical audiences, executive communication & presentation, teaching & curriculum delivery, workshop facilitation, mentorship & capacity building, cross-functional collaboration

Scaled my skills into my teams

Taught Electron to 50+ colleagues enabling follow-on projects; TA for a 60-student course; coached 40+ FIRST Robotics students.